

Curriculum vitae - Sariah Al Saati

Postdoctoral researcher in Topology in Condensed Matter Physics

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Education

- **Ph.D. in Physics**

Centre de Physique Théorique, École polytechnique, Palaiseau, France - 2022/2025

- Dissertation: Topological insulators and local properties of orbital magnetization
- Supervisors: Karyn le Hur and Frédéric Piéchon

- **Master of Science (M2) at International Center of Fundamental Physics**

École normale supérieure, Paris, France - 2021/2022 - Grade: 16.88/20

- Specialization: Fundamental quantum physics and theoretical condensed matter.
- Relevant coursework: Lie groups/algebras and representation, Differential geometry and gauge theory, Topology in condensed matter.

- **Engineering Degree (Ingénieur de l'École polytechnique - X18)**

École polytechnique, Palaiseau, France - 2018/2022 - GPA 3.87/4

- Major: Space engineering specialized on satellite control in orbit.
- Relevant coursework: Fundamental Physics (Quantum P., Relativity, Electromagnetism, Statistical P.), Mathematics (Optimization, Control, Statistics, Functional Analysis, Differential Calculus), Computer Science (Algorithm Design and Analysis, Fundamentals of Computer Science: Logic, Models and Computing).
- Various programming projects: Game dev., Web dev., Software dev. for drone swarm control, Algorithmic dev. for network communication optimization.

Research Experience

- **Research intern in astrophysics and planetology**

Institut de Recherche en Astrophysique et Planétologie, Toulouse, France - 2021/2022

- Subject: Magnetosphere-Ionosphere-Thermosphere Coupling study at Jupiter Based on Juno's First 30 Orbits and Modelling Tools.

- **DESTINY project manager - BEXUS 28 Balloon Experiment**

Swedish National Space Agency, Kiruna, Sweden - 2019/2020

- Subject: Detection of Earthquakes through a Stratospheric Infrasound study.

Teaching Experience

- **Master's Project Supervisor** (2024/2025): Supervised a 6-month research project investigating the spatial properties of gap modes in topological insulators.
- **Teaching Assistant for experimental soft matter physics** (2022/2025): Assisted in the experimental module on Soft Matter, guiding students through hands-on projects and experiments.
- **Volunteer Teaching** (2016/2026): Engaged in general teaching activities within social organizations, providing educational support and mentorship.

Publications

Journal publications

[1] **Sariah Al Saati**, Karyn Le Hur, and Frédéric Piéchon. Theory of local orbital magnetization: local Berry curvature. *ArXiv*, 2025. doi: 10.48550/ARXIV.2512.12343

[2] Noé Clément, Yuki Nakamura, Michel Blanc, Yuxian Wang, and **Sariah Al Saati**. Ionospheric conductances at the giant planets of the solar system: A comparative study of ionization sources and the impact of meteoric ions. *Journal of Geophysical Research: Space Physics*, 130(1), January 2025. ISSN 2169-9402. doi: 10.1029/2024ja033061

[3] Karyn Le Hur and **Sariah Al Saati**. Quantum hall and light responses in a 2d topological semimetal. *Comptes Rendus. Physique*, 25(G1):415–432, December 2024. ISSN 1878-1535. doi: 10.5802/crphys.202

[4] Jean-Eric Wegrowe, Luqian Zhou, and **Sariah Al Saati**. Power efficiency of hall-like devices: Comparison between reciprocal and antireciprocal onsager relations. *Phys. Rev. B*, 110:024412, Jul 2024. doi: 10.1103/PhysRevB.110.024412

[5] C. K. Louis, P. Louarn, B. Collet, N. Clément, **Sariah Al Saati**, J. R. Szalay, V. Hue, L. Lamy, S. Kotsiaros, W. S. Kurth, C. M. Jackman, Y. Wang, M. Blanc, F. Allegrini, J. E. P. Connerney, and D. Gershman. Source of radio emissions induced by the galilean moons io, europa and ganymede: In situ measurements by juno. *Journal of Geophysical Research: Space Physics*, 128(12), December 2023. ISSN 2169-9402. doi: 10.1029/2023ja031985

[6] Karyn Le Hur and **Sariah Al Saati**. Topological nodal ring semimetal in graphene. *Physical Review B*, 107(16), April 2023. doi: 10.1103/physrevb.107.165407

[7] **Sariah Al Saati**, N. Clément, C. Louis, M. Blanc, Y. Wang, N. André, et al. Magnetosphere-ionosphere-thermosphere coupling study at jupiter based on juno's first 30 orbits and modeling tools. *Journal of Geophysical Research: Space Physics*, October 2022. doi: 10.1029/2022ja030586

Conference Paper

[8] **Sariah Al Saati**, Elsa Deville, Nathan Vaneberg, Théo Boyer, Gatien Fonmartin, Louis Dubois, and David Mimoun. Detection of earthquakes through a stratospheric infrasound study - destiny, bexus 28. *Proc. 25th esa symposium on european rocket and balloon programmes and related research*, October 2022

Conferences and Communication

Conferences

- Sariah Al Saati et al., *Detection of earthquakes through a stratospheric infrasound study - destiny, bexus 28*. **25th esa symposium on european rocket and balloon programmes and related research**, Biarritz, May 2022 - Poster and 5 min talk, awarded best poster prize of the conference.
- Sariah Al Saati, Noé Clément, Corentin Louis, Michel Blanc et al., *Magnetosphere-Ionosphere-Thermosphere Coupling study at Jupiter Based on Juno First 30 Orbits and Modelling Tools*. **Magnetospheres of the Outer Planets meeting 2022** Liège, July 2022 - Poster and 5 min talk.
- Sariah Al Saati and Karyn Le Hur. *Light-induced topological semimetal in graphene*. **DFG FOR 2414 Annual Workshop 2023** Hamburg, March 2023 - 45 min talk.
- Sariah Al Saati and Karyn Le Hur. *Quantum Hall and Light Responses in a 2D Topological Semimetal* **QUANTUMatter 2024** San Sebastian, May 2024 - Poster.
- Sariah Al Saati, Valentin Desbuis, and Jean-Eric Wegrowe. *Reciprocal vs. antireciprocal Onsager relations: Hall devices as archetype* **Sadi Carnot's legacy** Palaiseau, September 2024 - Poster and conference staff member.
- Sariah Al Saati and Karyn Le Hur. *Quantum Hall and light responses in a 2D topological semimetal* **CT.QMAT24** Dresden, September 2024 - Poster.
- Sariah Al Saati, Frédéric Piéchon and Karyn Le Hur. *Thermodynamic theory of local orbital magnetization* **Mallorca Topological Quantum Matter** Palma, Mai 2025 - Poster.
- (Upcoming) Sariah Al Saati, Frédéric Piéchon and Karyn Le Hur. *Thermodynamic theory of local orbital magnetization* **SPIE Photonics Europe - Spintronics and Nanomagnetism** Strasbourg, April 2026 - 25 min talk.

Seminars

- Sariah Al Saati. *Introduction to topology in condensed matter* **CPHT Young Researcher Seminar** Palaiseau, January 2025 - 45 min talk.
- Sariah Al Saati, Frédéric Piéchon and Karyn le Hur. *Thermodynamic theory of local orbital magnetization* **Luxembourg theory group seminar** Luxembourg, November 2025 - 45 min talk.